Contract size and dynamics of stock index futures volatility and trading activity

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Abstract

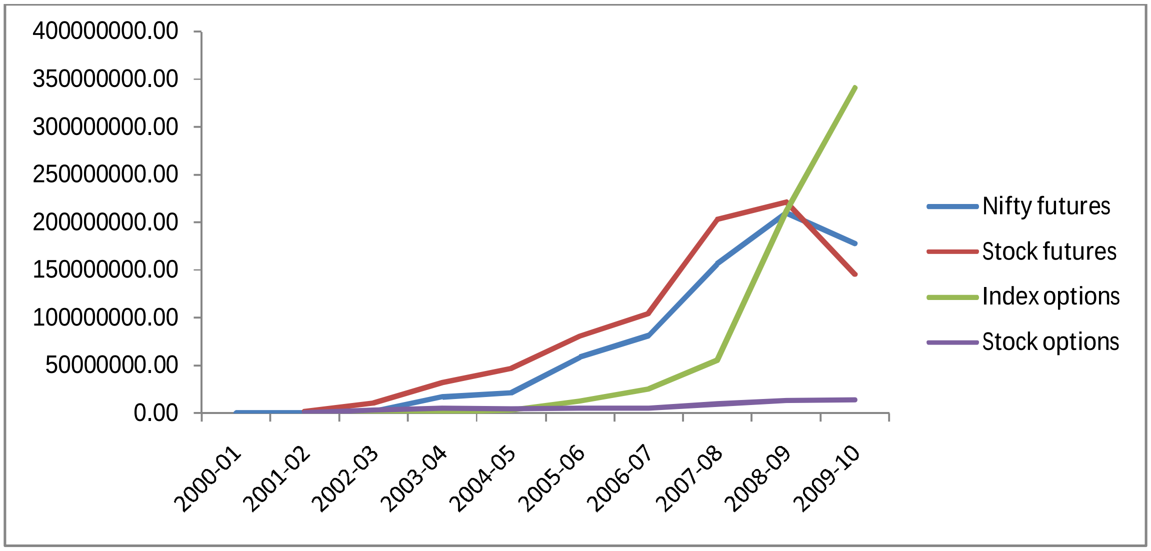
In an effort to increase the accessibility and attractiveness to the investors, futures exchanges alter the size of the futures contract. National Stock Exchange of India (NSE) had reduced Nifty stock index futures two times from 200 to 100 w.e.f 01/04/2005 and from 100 to 50 w.e.f 07/02/2007 to increase the trading activity. Does the Nifty futures live up to the expectation of the players supporting hedge based and information based trading following the reduction in contract size? We evaluate the impact of the reduction of the contract size on the use of the Nifty index futures contract by examining the dynamics of daily return volatility and trading activity variables such as volume and open interest through VAR model. We find that following the reduction in contract size both volume and volatility are increased and volume is caused by volatility , conforming the information based trading which would contribute to better price discovery. Further, volatility is caused by the open interest and not vice versa reinforcing the noise trading in the Nifty futures contract. Finally inconsistent in causality from volatility to open interest during the period of reduction, indicates hedge based trading does not depend on contract size of the stock index futures contract.

Keywords: VAR, Volatility, Stock index, Granger causality, Volume and Open interest

Introduction

A review of the relationship between futures price volatility and trading volume is the key to developing countries since the volume volatility relationship has a greater role in designing innovative futures contracts. Further, a better understanding of volume volatility relationship in the futures market is important for application in the context of asset pricing and portfolio insurance.

Figure – 1 Product wise no of contracts traded in NSE derivative segment from June 2000 to September 2009



In the figure -1, the Nifty index futures stands out as the second largest contract trading in National Stock of Exchange of India (NSE) only after all the stock futures taken together in terms turnover, apparently showing a steady growth in the trading activity in Nifty futures. Is it due to information based trading or hedge based trading or due to noise trading in the futures contract? Further, during the sample period of the study the lot size of the contract reduced two times from 200 to 100 and further to 50 units. How does this change in lot size impact the dynamics of nifty futures volatility and its trading activity variables. The study tried to go deep into the informational structure of Nifty index futures contract following the two consecutive reductions in the contract size. Moreover, as a popular equity derivative instrument in the market, Nifty index futures is used by hedgers, speculators and arbitrageurs. Does the Nifty futures contract come up to the expectation of the players supporting hedge based and information based trading? The importance of trading activity variables also lies with the return predictability in the futures market. Wang and Yu (2004) studying 24 US futures market, found that short term profits on a contrarian portfolio (buying past losers and selling past winners) can be predicted from past trading activity variables such as volume and open interest. Since lag volume and lag open interest share a positive and negative relationship respectively with the current profit, higher volume and lower open interest futures contract will give better contrarian profit.In addition, emerging market is characterized by higher persistence of return and volatility symbolizing inefficiency offering the opportunity for excess risk adjusted return. In their comparative study of the performances of passive and active trading strategy on Taiwan stock index futures, active index futures trading strategy outperform the passive i.e. buy and hold strategy, Chiang et.al.(2012).

The underlying Nifty 50 stocks index is the bellwether index representing the 22 sectors of the economy and 63% of the market capitalisation of the National Stock Exchange of India giving compounded annual growth rate of 13.13% from 2000-01 to 2009-10, being the third largest exchanges by number of stock index future contract traded in 2009 after CME group and Eurex.[1]

The stock exchanges try to design optimal size of futures contract to enhance the attractiveness and accessibility of the contract to achieve the two most important economic functions of derivative i.e. risk management and price discovery, besides enhancing the liquidity, because large contract size keeps the small traders away from trading in futures market due to capital constraint and makes the hedgers unable to fine tune their hedging program with respect to their spot market commitment, Silber (1981). Lower transaction cost, leverage and liquidity of the futures market lead to efficient and faster dissemination of information in the futures market, leading to the information based trading in futures implies the price discovery function of futures market (Zhou et al, 2014). Furthermore, it is argued by the future market advocates that information on subsequent spot price movement can be known from the futures market helping the players in managing the risk they are exposed to in the spot market, Darrat & Rahaman (1995) , Pericli & Koutmos (1997) and Darrat et.al. (2002)

Following the reduction in contract size, increase in trading activities is observed in terms of higher adjusted volume and open interest. Further, trading frequency is also increased which led to higher price volatility (Bjursell et al. 2010). Greater liquidity in terms of lower spread and higher trading volume is documented because of decrease in contract size, Karagozoglu and Martell (1999). However, the study by Huang and Stoll (1998) predicted an increase in

[1] Indian Security Market – A Review, Volume XIII 2010, www.nseinda.com

liquidity and smooth in price fluctuation following the decrease in contract size. They reported that the reduction in contract size allowed the new small investors to enter the market who previously might have capital constraint due to higher contract value. Further, motivated by this study , Karagozoglu et al (2003) studied the impact on price volatility of the reduction in contract size of S&P 500 trading in Chicago Mercantile Exchange (CME) and reported no significant impact. However their result might have been confounded by the de-facto reduction in contract size which refers to the introduction of E-mini S&P 500 futures contract on CME in September 1997 during the period of the study.

Amid the inconclusive findings of the impact of change in contract size on liquidity and price volatility, Nifty index futures is the ideal market for studying the impact of lot size changes on the dynamics of volatility and trading activity variables. The descriptive statistics of the trading activity variables such as volume and open interest and conditional volatility are reported in the Table -1 for the three different sub-periods of the study in which each period represents a trading in particular contract size following the change.

Table -1

Descriptive statistics of trading volume, open interest and volatility

	Contract Size 200					**Contract Size 100**					**Contract Size 50**				
	GARCH01	OI_100	OI	VOL_100	VOLUME	GARCH01	OI_100	OI	VOL_100	VOLUME	GARCH01	OI_100	OI	VOL_100	VOLUME
Mean	0.000253	1.017575	3629920	1.035022	104403.2	0.000267	1.004925	22075967	1.007433	671516.7	0.000605	0.999544	29714684	1.003027	1277367
Median	0.000168	1.016775	1893000	1.026211	16587.18	0.000165	1.009936	22981500	1.005801	663118.5	0.000461	1.003187	30927250	1.002118	1244392
Maximum	0.006006	1.134429	17198400	1.310481	720437.5	0.002151	1.035065	37084200	1.086502	1863217	0.004689	1.027973	44400050	1.078559	3091293
Minimum	6.19E-05	0.888918	4000	0.763283	51.87	6.90E-05	0.9321	5534900	0.849056	101241.3	7.54E-05	0.923456	7272150	0.851591	131141.5
Std. Dev.	0.000395	0.033931	4006149	0.065963	144365.5	0.000309	0.01961	5723029	0.028013	312250.1	0.000593	0.01786	7467872	0.023086	399724.7
Skewness	9.051996	0.490164	1.136981	0.686921	1.380761	3.440982	-1.60552	-0.680226	-0.39331	0.732384	3.175952	-1.44672	-0.6968	-0.36729	0.84816
Kurtosis	107.4346	4.624241	3.14451	4.896333	4.155072	15.68247	5.610481	3.683959	6.1528	3.86036	16.35718	5.397237	3.180091	7.068493	4.869162
Jarque-Bera	565462.3	166.4632	261.5364	253.6128	451.3695	4025.327	331.0909	44.82682	204.1391	55.79149	5924.773	382.3825	53.47756	462.9147	172.5553
Probability	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Sum	0.30524	1129.508	4.39E+09	1148.875	1.26E+08	0.123984	466.2851	1.02E+10	467.449	3.12E+08	0.393376	649.7037	1.93E+10	651.9674	8.30E+08
Sum Sq. Dev.	0.000188	1.276774	1.94E+16	4.825373	2.52E+13	4.42E-05	0.17804	1.52E+16	0.363334	4.51E+13	0.000229	0.207019	3.62E+16	0.345893	1.04E+14
Observations	1208	1110	1209	1110	1209	464	464	464	464	464	650	650	650	650	650

It is clearly evident from the descriptive statistics that there is increase in trading activity such as volume and open interest following the reduction in contract size from a multiple of 200 to 100 and finally to 50. Also conditional volatility has been increased during those periods followed by the reduction in contract size. These market episodes indicate a possible well-built interrelationship among the sign and scale of price movements, volatility of prices and the trading volume as shown in many previous studies (Gallant et al. 1992).

The study on price-volume relations in financial markets has long been an interesting phenomenon for many researchers in financial economics. The section of finance literature dedicated to the study of the relationship between the volatility, return and trading volume of futures contracts tried to explore the information arrival dynamics in the financial market there by providing evidence about either of the four complementary information theories, namely, the mixture distribution hypothesis (MDH) advocated by Clark (1973), sequential information arrival hypothesis (SIAH) by Copeland (1976), dispersion of belief hypotheses, Shalen (1993), Harris and Raviv (1993) and noise trading hypotheses, Delong,Shleifer, Summers, and Waldmann (1990a, 1990b). As far as futures market is concerned, the study of the informational dynamics of how trading activity variables such as volume and open interest and volatility holds the importance for the futures market traders' i.e. hedgers, speculators and arbitrageurs. The theories not only diverge in terms of pragmatic relationship between price changes and trading volume, but also they are inconsistent across market and products. The paper tries to provide an integrated picture by investigating the short term linear dynamics of the association between volume and volatility in Nifty index futures contract along with open interest, another important trading activity variable in the futures market.

According to the MDH, the relationship between asset price volatility and trading volume is a function of a common latent variable, that is, the rate at which information arrives in the market (Clark1973; Epps and Epps 1975; Harris 1984). To put it simply, new information in the market is an underlying variable that directs a contemporaneous or simultaneous change in the price and the trading volume, thus creating a positive relationship without causation between price and volume (Clark 1973).However, the explanation for a causal relationship between asset prices and trading volume relies on the SIA hypothesis, according to Copeland (1976) and later extended by Jennings et.al. (1981) and Smirlock and Starks

(1984). This hypothesis states that information dissemination occurs sequentially to investors, causing a series of intermediate equilibrium prices and thus leading to a final informational equilibrium price when all the investors are informed. Therefore, this hypothesis is construed to be more consistent since it not only advocates contemporaneous change but also talks about the lagged relation between volume and volatility. Because of traders' sequential response to the arrival of new information, the price achieves an informational equilibrium until the information reaches all traders. Thus, a scenario emerges in which a lead-lag relationship between trading volume and volatility is established.

Confirming the Wall Street wisdom "It takes volume to make prices move, "a major portion of studies broadly acknowledged a positive contemporaneous correlation between price volatility and trading volume of financial assets in the cash market (Karpoff 1987). Despite the sweeping documentation of the positive relationship between trading volume and price changes in the spot market, there is no clear-cut evidence; instead, mixed results have been reported for the futures market. Empirical investigation of this relationship in the futures market began in 1973 with Clark's pioneering study of the cotton futures market; Clark (1973) found that the square of price changes shares a positive relationship with aggregated trading volume. Subsequently, many studies concentrating either on commodity futures, Treasury bill futures or foreign currency futures have revealed a positive relationship between trading volumes and return variability (Cornell 1981; Tauchen and Pitts 1983; Grammatikos and Sunders 1986).

Concerning stock index futures, in a seminal study Kawaller et al. (1987) examined the relationship between volume and futures prices by using a simultaneous equation model on the transaction data for the S&P 500 index futures from 1984 to 1986. Paralleling other studies in the commodity and currency markets, this study also revealed a significant positive impact of daily trading volume on the volatility of S&P futures contracts. With a deviation in methodology, Board and Sutcliffe (1990) used Pearson partial correlation coefficients to examine the degree of daily movement in volume and volatility. The authors also confirmed a positive association between volume and volatility even when the weekend effects were removed. Kutan and Aksoy (2003), even after taking into account the inflation and export adjusted return, documented the importance of public information arrival, represented by volume in the study, in the Turkish asset market. Mougoue and Aggarwal (2011), in a study on three major dollar-denominated currency futures contracts using linear and nonlinear Granger causality models, reported a significant lead-lag relation between trading volumes and return volatility, consistent with the sequential arrival of information hypothesis.

However, Kalotychou and Staikouras (2006), taking trading volume and open interest as variables representing liquidity, documented a significant but weak, as a conduit of transmitting information into the market in a Generalized Autoregressive Conditional Heteroskedasticity (GARCH) framework.

The paper is motivated by the prior inconclusive literature on relationship dynamics between volatility and trading activity variables across product and markets and the impact of the change of the contract size on this dynamics. Therefore, attempt has been made in this study to find out the impact of the change in the lot size of the Nifty futures contract on the dynamics of its volatility and trading activity.

Optimal size of the futures contract is one of the important determinants of the trading by speculators (informed and noise traders) and hedgers Silber (1981). The objectives of the study is to find out whether the dynamics of the Nifty futures contract has become more efficient as an instrument of risk management and price discovery; the two most important economic functions of derivative following the reductions in the contract size of the index futures contract. It was found that, other than fundamental factors, exogenous factors like regulatory structure of the market impact the relationship dynamics amongst price volatility, volume and open interest, (Watnabe, 2001). Does small lot size bring more information to the market leading to better price discovery and as an instrument for better risk management, invite more players to hedge? If not, speculative activity will bring about noise trading due to small lot size which will destroy the whole economic purpose of index futures contract.

In order to prove the objectives of the study, the following three hypotheses were tested in this paper.

H-10: Hedge based trading – Following the reduction in contract size more hedging activities are happening as small size of the contract will enable the hedgers to fine tune their hedging program. It reinforces the risk management function of the derivative contract.

H-20: Information based hedging- More informed traders will trade due to small size of the futures contract supplemented by the fact that the futures market is highly leveraged and transaction cost is lower. It ensures the price discovery function of the derivative product.

H-30: Noise based trading- More noise trading by uninformed investors being induced by small contract size. Due to reduction in lot size more small traders who are uninformed are interested to enter the market, Bjursell et.al. (2010), creating more price volatility which is opposite to the price discovery function of the futures contract.

2 Data and methodology

We used the daily closing prices, trading volume (turnover) and open interest of the near month futures contract from 6 June 2000 (the date on which futures contracts on CNX Nifty Index started trading on the NSE) through 29 September 2009. In total, we have 2,323 observations of closing price, trading volume and open interest obtained from the official website of the National Stock Exchange of India (NSE).[2] CNX Nifty is a bellwether index of the NSE representing 50 stocks across 22 sectors. The NSE is the world's fourth largest exchange in number of trades in equity and one of the largest in terms of equity derivatives volume globally. Nifty represents 52% of the traded volume of the NSE and 63% of the NSE's market capitalization. NSE, in order to attract more market participants to the Nifty futures contracts, reduced the lot size from 200 to 100 and 100 to 50 units w.e.f April 1 2005 and Feb 7, 2007 respectively, two times in the sample period. Accordingly, data have been divided into three subsamples considering the no of split of the index futures in the study period. In the first sub sample the period spans from 06/12/2000 through 03/31/2005 with the lot size of 200 multiples of Nifty index. The second sample period starts from 04/01/2005 to 02/06/2007 with 100 multiples of the Nifty index futures and finally lot size of 50 from 02/07/2007 to 09/24/2009.

Following Campbell et.al (1993) and Fund & Patterson (1999), for the study we have formed stationary time series of trading volume and open interest by incorporating the following procedure through Eq. (1) using 100 days backward moving average. The resultant time series captures the change in the long run movement in trading volume and open interest.

$$V_t = \frac{VT_t}{\frac{1}{100}\sum_{i=1}^{100} VT_{t-i}} \qquad (1)$$

Where, VT_t is the log of trading volume at time t and for open interest it represents the log open interest at time t. The transformation through Eq. (1) helps to mitigate the mechanical link between volume and open interest in the actual trading and enables to interpret the information structure relating to volume, open interest and price movements through VAR

[2] www.nseindia.com

analysis. The daily return series are calculated as the first difference of the logarithms of daily closing prices of near month Nifty index futures contracts.

The unconditional daily mean return of the Nifty index futures is 0.000531%, with a standard deviation of 0.018%. The return series is fat-tailed as reported by excess kurtosis of 11.53 and negatively skewed since the reported skewness is –0.48. The Jarque-Bera test rejects the null hypothesis of normality (available on request). The evidence of the existence of excess kurtosis is due to a possible time varying variance, as documented by Akigary et.al. (1991) and Fujihara and Mougoue (1997). Further, the presence of heteroskedasticity in the index return is significant at lag 10 as per the ARCH LM test, Engle (1982). Therefore, conditional volatility of Nifty index futures is taken for the study of the dynamic interrelationship among volatility, volume and open interest in near month future contract. The ARCH class of model i.e. GARCH (1, 1) being parsimonious is used to estimate the conditional volatility of return. Conditional volatility following GARCH (1, 1) model proposed by Bollerslev (1986) is estimated. In GARCH (1, 1), the conditional variance h_t is a function of a long run average variance rate , lagged squared error term represented by news and its own lag.

$$r_t = c + br_{t-1} + \varepsilon_t \quad (2)$$

$$\varepsilon_t / I_{t-1} \sim N(0, h_t) \quad (3)$$

$$h_t = \omega + \alpha\varepsilon^2_{t-1} + \beta h_{t-1} \quad (4)$$

Augmented Dicky-Fuller (ADF; 1979) test is conducted to test the presence of unit root in the return volatility series (h_t), volume and open interest. The ADF test results[3] strongly reject the null hypothesis that the Nifty index futures return volatility series and trading volume and open interest series are non-stationary. Therefore, all three time series are taken for further analysis in this study.

Methodology

To investigate the interrelationship between Nifty index futures volatility, volume and open interest in the near month contract, Vector Autoregressive (VAR) model is used in the study. It is the generalized multivariate form of univariate autoregressive model, carrying the most valuable feature of estimating the dynamic interrelationship between variables without any choice of having dependent and independent variables. Further, Granger causality is conducted by estimating the VAR model to determine the causal relationship

[3] Results are available on request.

among Nifty index futures volatility and trading activity variables such as volume and open interest. Causal relationship does not imply any cause and effect relationship, rather it is based on “predictability or forecastabilty”, Granger (1969). Further going deep into the nature of forecastability exists between the variables; the study estimates the variance decomposition to find out magnitude of the dynamic effect and impulse response function to trace out the direction and speed of the dynamic effect among the volatility, volume and open interest of Nifty index futures. Variance decomposition helps in finding the variation in each variable in the system that is contributed by its own shock and shocks produced from other variables. The impulse response function provides the direction and time profile of the effects of own shocks and shocks from other variables in the system on the future behavior of the variables.

Expressing in its reduced form, the VAR model used in the study is presented as follows to analyses the interrelationships among daily volatility, trading volume and open interest on Nifty index futures near month contract.

$$Y_t = c + \sum_{k=1}^{L} a_k \, Y_{t-k} + bM + e_t \qquad (5)$$

Where, Y_t is a 3×1 column vector of volatility (GARCH01), trading volume (VOL) and open interest (OI) at time t. c and *ak* are 3x1 and 3×3 matrices of coefficients, L is the lag length , e_t is the 3x1 column vector of error terms. The direct effect of *i*th variable upon *j*th variable after k periods is measured by the (*i, j*) component of α_k . Specifically, the *i*th component of e_t is the innovation of the *i*th variable that cannot be predicted from the other variables in the system.

The VAR is rewritten in the form of following individual equations to test the granger causality to test the joint hypothesis that, “Do changes in *VOL* cause changes in *GARCH01”*?, likewise for other joint hypothesis for causality from OI to VOL, GARCH01 to OI, OI to GARCH01, and VOL to OI. As all the variables are in the VAR system are stationary their joint hypothesis can be tested using F-statistics.

Our first null hypothesis, H20, that no information based trading following the reduction in lot size i.e. there is no granger causality from volume to volatility and vice versa is expressed as:

$\beta_{21} = \beta_{22} =$ ------------------- $= \beta_{2L} = 0,$ *and*

$\gamma_{11} = \gamma_{12}$ --------------------- $= \gamma_{1L} = 0$

A rejection of the joint hypotheses of zero value lagged β_{2k} and γ_{1k} supports the information based trading in Nifty futures.

The second null hypotheses, H10, that no hedge based trading following the reduction in lot size i.e. there is no granger causality from volatility to open interest is expressed as:

$\delta_{11} = \delta_{12} =$ --------------- $= \delta_{1L} = 0$

A rejection of the joint hypotheses of zero value lagged δ_{1k} supports the hedge based trading in Nifty futures.

The third hypothesis, H30, that no noise trading following the reduction in lot size i.e. there is no granger causality from open interest to volatility is expressed as:

$B_{31} = \beta_{32} =$ ------------------ $= \beta_{3L} = 0,$

A rejection of the joint hypotheses of zero value lagged β_{3k} supports the noise trading in Nifty futures.

$$\mathbf{GARCH01} = c_{10} + \beta_{1k} \sum_{k=1}^{L} \mathbf{GARCH01}_{t-k} + \beta_{2k} \sum_{k=1}^{L} VOL_{t-k} + \beta_{3k} \sum_{k=1}^{L} OI_{t-k} + e_{1t} \tag{6}$$

$$\mathbf{VOL}_t = c_{20} + \gamma_{1k} \sum_{k=1}^{L} \mathbf{GARCH01}_{t-k} + \gamma_{2k} \sum_{k=1}^{L} VOL_{t-k} + \gamma_{3k} \sum_{k=1}^{L} OI_{t-k} + e_{2t} \tag{7}$$

$$\mathbf{OI}_t = c_{30} + \delta_{1k} \sum_{k=1}^{L} \mathbf{GARCH01}_{t-k} + \delta_{2k} \sum_{k=1}^{L} VOL_{t-k} + \delta_{3k} \sum_{k=1}^{L} OI_{t-k} + e_{3t} \tag{8}$$

In an effort to provide further insight into the dynamic interrelationship among the variables, VAR model is represented as moving average model of innovations of the system variables.

$$Y_t = \sum_{k=1}^{\infty} A_k e_{t-k} \tag{9}$$

The m step ahead forecast error of Y_t can be computed from the moving average model of VAR in Eq. (9). Breaking up of the variation of the forecast error of the variable $Y_{t,}$ variance decomposition explains the unexpected change in the variation of each variable produced by shocks from other system variables.

Impulse response function traces out the responsiveness of the dependent variable in the system to the shocks /innovations or impulses of each variable in the system.

Eq. (9) is transformed and presented as follows to remove the contemporaneous correlation across equation by the process of orthogonalization. Though the innovations e_t is serially uncorrelated , but across equations in the VAR system, they might be correlated.

$$Y_t = \sum_{k=1}^{\infty} A_k V u_{t-k} \qquad (10)$$

Such that,

$$Y_t = \sum_{k=1}^{\infty} C_K u_{t-k} \qquad (11)$$

V is a lower triangular matrix.

Now μ_t is serially and contemporaneously uncorrelated innovations which are orthogonalized innovations from e= Vμ with an independent covariance matrix.

Empirical Analysis

In this section the results of VAR analysis is reported. The significant estimates of VAR (available on request) and impulse response function in Figure (2, 3,and 4) of volume, open interest and volatility following the decrease in contract size from 200 to 50 , the variables are significantly impacted by their own lags. The response of volatility to its own shock became highly persistence than prior to the change in lot size. The impact does not die out in small lot while in larger lot it dies out within next 20-21 days of trading to the shock. As far as trading activity variables are concerned, the impact of their own shock is less persistence following the reduction in lot size. In case of open interest the impact dies out within 11 days and for volume within 4 days of the shocks. However, prior to the change with lot size of 200 the impact of the innovation never dies out in the life of the contract. It signifies inefficiency due to lack of liquidity.

Table : 2 – Granger causality of open interest (OI), Volume (VOL) and return volatility (GARCH01) , Chi-sq values with p values.

Dependent variable: OI	**Contract Size 200**			**Contract Size 100**			**Contract size 50**			**Full Period**		
	Chi-sq	df	Prob.	Chi-sq	df	Prob.	Chi-sq	df	Prob.	Chi-sq	df	Prob.
VOL	13.13644	2	**0.0014*** **	4.73782	1	**0.0295** **	2.690562	1	**0.1009***	23.82611	2	0.000***
GARCH01	2.931166	2	0.2309	3.754203	1	**0.0527** **	0.581999	1	0.4455	6.635211	2	0.0362
Dependent variable: VOL												
	Chi-sq	df	Prob.	Chi-sq	df	Prob.	Chi-sq	df	Prob.	Chi-sq	df	Prob.
OI	16.97928	2	**0.0002*** **	2.053432	1	0.1519	3.210704	1	**0.0732***	41.65201	2	0.000***
GARCH01	3.33855	2	0.1884	0.001742	1	0.9667	5.141418	1	**0.0234** **	11.05285	2	0.004**
Dependent variable: GARCH01												
	Chi-sq	df	Prob.	Chi-sq	df	Prob.	Chi-sq	df	Prob.	Chi-sq	df	Prob.
OI	2.939719	2	0.23	1.192222	1	0.2749	5.440987	1	**0.0197*** **	4.761144	2	0.0925*
VOL	3.563612	2	0.1683	30.75514	1	**0.000*** **	5.402987	1	**0.0201** **	0.100914	2	0.9508

***, ** and * indicate 1%, 5% and 10% level significance

Table : 3 Variance decomposition of open interest (OI), volume (VOL) and volatility (GARCH01)

Variance Decomposition of OI:	Lot size 200			Lot Size 100			Lot Size 50			Full Period		
Period	OI	VOL	GARCH01	OI	VOL	GARCH01	OI	VOL	GARCH01	OI	VOL	GARCH01
1	100	0	0	100	0	0	100	0	0	100	0	0
2	99.6081	0.315831	0.076066	99.40238	0.561178	0.03644	99.78208	0.20773	0.010189	99.84855	0.096848	0.054603
3	99.07253	0.733532	0.193934	98.64209	1.240854	0.117052	99.5159	0.456755	0.027349	99.37154	0.488432	0.140029
4	98.41976	1.247472	0.332768	97.95949	1.807119	0.23339	99.29424	0.658472	0.04729	98.77604	0.965753	0.258202
5	97.73879	1.780498	0.480714	97.40527	2.220761	0.373973	99.1347	0.797812	0.067484	98.10537	1.49399	0.400641
10	94.82833	3.995623	1.176048	95.85343	3.012653	1.133918	98.86565	0.991756	0.142598	95.11723	3.593048	1.289722
15	93.18963	5.163455	1.646916	95.07984	3.213074	1.707087	98.82376	1.000318	0.175922	93.38266	4.491713	2.125625
20	92.40817	5.686018	1.905814	94.59981	3.314585	2.085609	98.81044	1.000417	0.189141	92.49769	4.783704	2.718602
25	92.0589	5.908217	2.032879	94.28998	3.377928	2.332094	98.80533	1.000367	0.194306	92.05537	4.865048	3.079582
Variance Decomposition of VOL:	Lot size 200			Lot Size 100			Lot Size 50			Full Period		
Period	OI	VOL	GARCH01	OI	VOL	GARCH01	OI	VOL	GARCH01	OI	VOL	GARCH01
1	4.259531	95.74047	0	0.259593	99.74041	0	3.154539	96.84546	0	3.288499	96.7115	0
2	5.213807	94.70536	0.08083	0.648644	99.35134	1.95E-05	4.109535	95.78508	0.105381	4.009674	95.74886	0.241471
3	6.519668	93.28889	0.191442	0.983726	99.01624	3.08E-05	4.790027	94.90141	0.308568	5.568015	94.07333	0.358658
4	7.782213	91.8732	0.344591	1.205698	98.79409	0.000217	5.204447	94.23677	0.558779	7.110348	92.37403	0.51562
5	9.009908	90.47816	0.51193	1.331221	98.66806	0.000724	5.427141	93.7577	0.815163	8.638823	90.69059	0.670583
10	13.62235	85.09877	1.27888	1.447322	98.5458	0.006879	5.586792	92.68026	1.732948	13.9708	84.5235	1.505698
15	15.9141	82.3808	1.705095	1.448398	98.53827	0.013337	5.566449	92.31518	2.118368	16.0864	81.67979	2.233813
20	16.92157	81.18407	1.89436	1.448402	98.53372	0.017877	5.561529	92.16976	2.268707	16.77719	80.48008	2.742732
25	17.34733	80.67949	1.97318	1.448469	98.53065	0.02088	5.560474	92.11237	2.327151	16.97797	79.97062	3.051417
Variance Decomposition of GARCH01:	Lot size 200			Lot Size 100			Lot Size 50			Full Period		
Period	OI	VOL	GARCH01	OI	VOL	GARCH01	OI	VOL	GARCH01	OI	VOL	GARCH01
1	0.072911	0.000821	99.92627	0.012859	0.807053	99.18009	0.102159	4.167705	95.73014	0.117794	0.233147	99.64906
2	0.058437	0.126129	99.81543	0.0774	5.186226	94.73637	0.08643	2.979411	96.93416	0.066055	0.232776	99.70117
3	0.095739	0.163677	99.74058	0.250652	8.905565	90.84378	0.194482	2.307594	97.49792	0.058351	0.255765	99.68588
4	0.147742	0.196737	99.65552	0.471204	11.57802	87.95078	0.353632	1.90987	97.7365	0.059962	0.283718	99.65632
5	0.204363	0.221845	99.57379	0.699792	13.43316	85.86704	0.52499	1.66244	97.81257	0.06556	0.31248	99.62196
10	0.489137	0.310511	99.20035	1.54837	17.21483	81.2368	1.173037	1.213984	97.61298	0.115441	0.437012	99.44755
15	0.689972	0.361721	98.94831	1.943844	18.27433	79.78183	1.455675	1.108752	97.43557	0.169789	0.519154	99.31106
20	0.796793	0.388063	98.81514	2.134001	18.73139	79.13461	1.565546	1.074149	97.3603	0.211703	0.567597	99.2207
25	0.84656	0.400403	98.75304	2.235585	18.97107	78.79335	1.607813	1.061461	97.33073	0.238843	0.594091	99.16707

Fig. 2 Impulse response of open interest to shocks in volume and volatility for contract size starting with 200, followed by 100, 50 and finally for the full period.

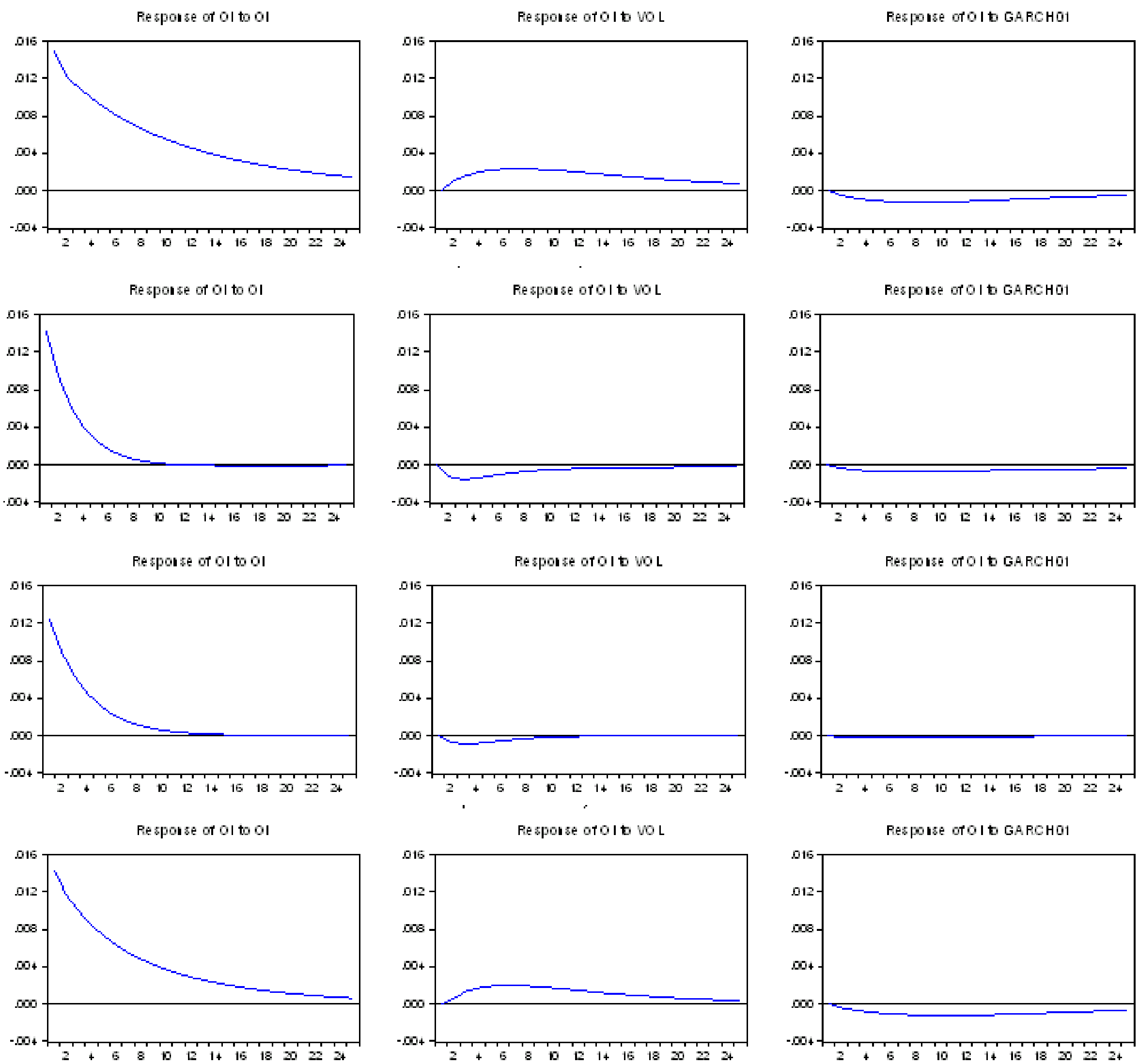


Fig. 2 Impulse response of volume to shocks in open interest and volatility for contract size starting with 200, followed by 100, 50 and finally for the full period.

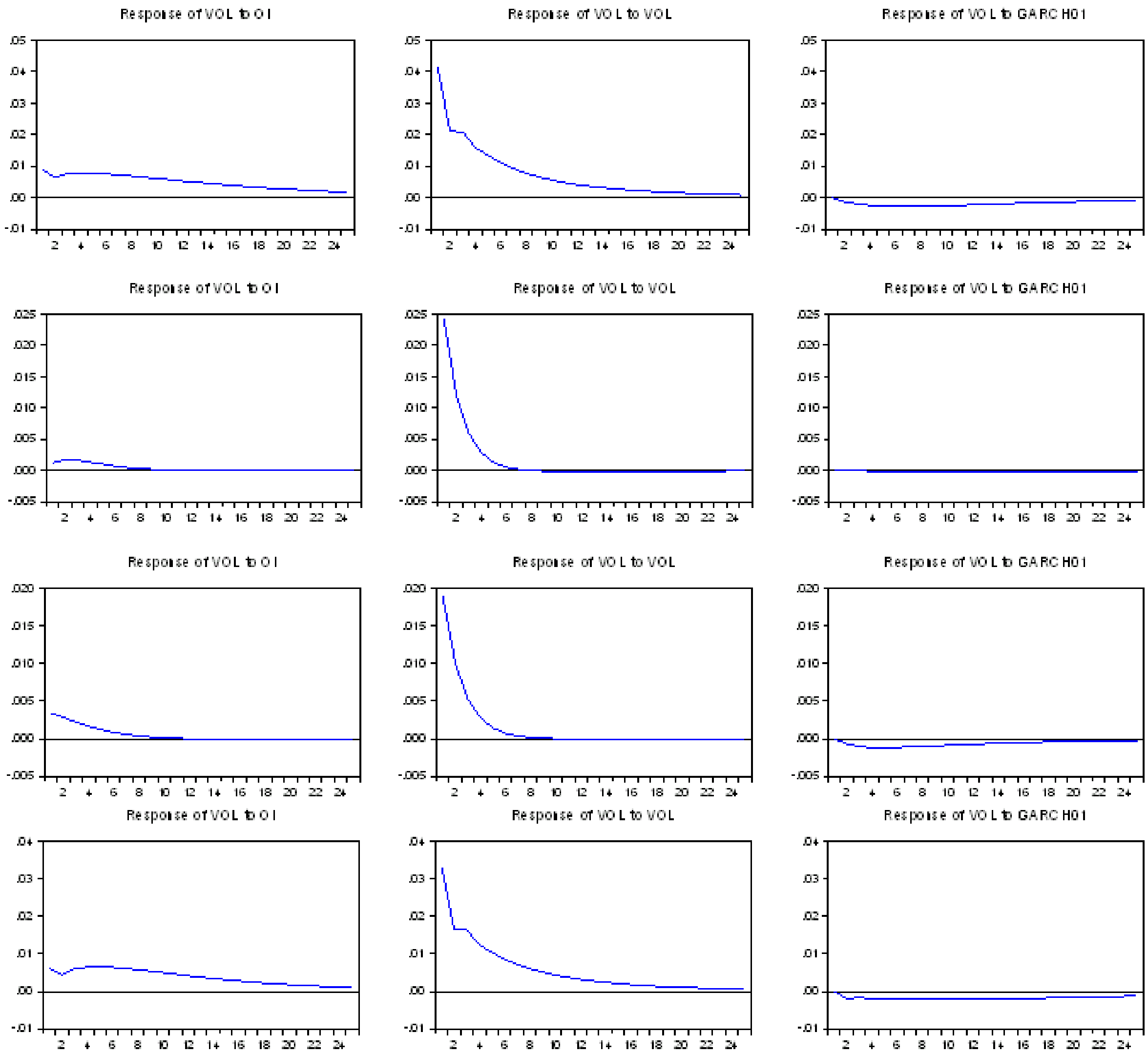
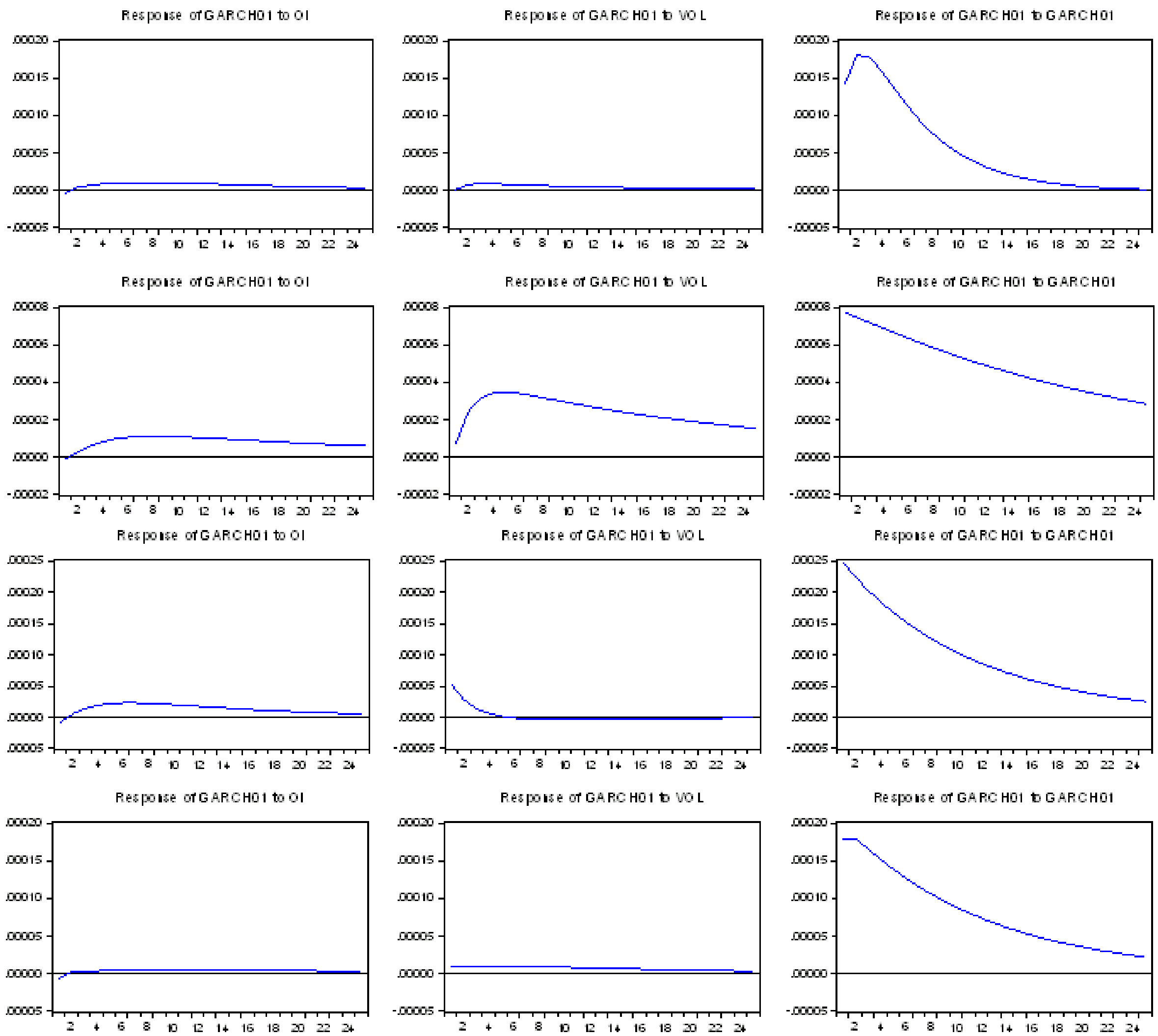


Fig. 2 Impulse response of volatility to shocks in volume and open interest for contract size starting with 200, followed by 100, 50 and finally for the full period.



Volume and volatility

Following the reduction of the contract size of the Nifty index futures, it resulted into information based trading. Prior to the reduction, neither volume is granger caused by volume, nor vice versa, whereas in post reduction period both volume and volatility share the feedback relationship. It is supplemented by the impulse response function of volatility to one unit positive shock to volume. The unexpected fluctuation in volume leads to higher volatility, but its impact dies out in 5 days. That means. With the small lot size of the Nifty futures contract, better price discovery is reported in relation the higher lot size which was leading to the persistence of the impact throughout the life of the near month futures contract. Thus given the information on previous day's volume, volatility can be predicted, thereby supporting the SIAH.

Though the feedback relationship reported between volume and volatility, the lead lag relationship is strong from volume to volatility, because around 4% of the unexpected variation in volatility is explained by the lagged volume. Also its dynamics is reported in the impulse response function of volatility from volume shocks. The one standard deviation of positive volume shock on volatility dies out in 4 days for smaller lot size of the index futures , where as it was highly persistent prior to the reduction. Its understood that following the reduction in lot size price discovery is very fast in Nifty index futures market, thereby supporting the information based trading hypotheses and reinforcing the theory of Sequential Information Arrival Hypotheses.

Further, the chi-sq statistics of granger causality from volatility to volume is significant at 2% level. It confirms the findings of Chen and Daigler (2008) who found that information reaches the different group of traders at different times. Also it reinforces the conventional presumption that price volatility is an efficient indication of information than volume of trade which is noisier, Shalen (1993) and Pfleiderer (1984). Moreover, the mean value of volatility rose from .000253 to .000605 (see Table -1) following the reduction , is driven by information, as the first moment of volatility process is driven by information and the second moment of volatility i.e. excess volatility is attributed to different interpretation of information, Shalen (1993)

Open interest and volatility

Unusual /unexpected open interest and volatility is observed following the decrease in contract size as reported in descriptive statistics, Table -1, the unconditional standard deviation in of both open interest and volatility increased. Taking open interest as a proxy for uninformed trading, liquidity trading and dispersion of belief, Bessembinder et. al. (1996), it is evident that though information reaches the traders at the same time, as with macroeconomic announcements, but its interpretation is different for different traders. Thus this unusual variation in open interest attributed to dispersion of belief in futures market hypothesis by Shalen (1993), which is generalized by Harris and Raviv (1993), caused unusual volatility. As reported in Table -2, open interest granger causes volatility and not vice versa, it does not support hedge based trading in the Nifty futures contract following the lessening of the contract size thereby rejecting our second hypotheses of the study.

Analyzing the feedback of the trading activity of the uninformed traders Chen and Daigler (2008), mispricing in stock index futures created by noise trading as a consequence to the dispersion of belief gets corrected by the informed traders responding via arbitrage. It is evident in the impulse response function of volatility to open interest shock, volatility declined following the reaction to the open interest shock and the impact of the shock gets corrected in the second day of trading. It is consistent with the literature on noise trading in stock markets, Delong et. al. 1990a, 1990b that uninformed traders are prevalent in the index futures market following the cutback in the lot size. So following the decrease of contract size, more number of uninformed trading creates noise in the Nifty futures, thereby rejecting our second hypotheses of more hedged based trading is happening in contract. However, with lot size of 100, open interest is granger caused by volatility and but not with lot size of 50, such change in the nature of relationship conforms that hedging does depend on the lot size of the contract

Volume and Open interest

Both volume and open interest are complimentary to each other as is evidenced from feedback relationships from the significant granger causality runs from volume and open interest and vice versa. However, the lead lag relationship is stronger from open interest to volume because 3-5% of the one day ahead forecasted volume explained by open interest and almost 0% for vice versa as reported through variance decomposition in **Table-3.** Furthermore, such stronger lead lag relationship from open interest to volume is strengthened by impulse response function of open interest. It is shown that open interest creates higher volume but the impact of the shock dies out within 8 days in **Figure -2**. But Nifty futures at its original lot size i.e. 200 the impact of the shock persists through the life of the contract. It substantiates the fact that in Nifty index futures besides information trading, non information trading is also happening. However, the noise created by the non information trading dies out with in the 8 days of the shock. Again the speed of adjustment information into the price in smaller lot size of the contract is faster than the larger lot. It supports our third hypotheses reduction in the contract size leads to noise trading in stock index futures.

6 Conclusions

The nature and dynamics of the relationship between return volatility and trading activity variables such as volume and open interest of Nifty index futures following the change in lot size have been studied through granger causality and impulse response function of VAR estimates. The feedback relationship between volatility and volume signifies the information based trading leading to the better price discovery. The one way causality from open interest to volatility confirms noise trading in Nifty futures contract in the smaller lot which is further strengthened by the feedback relationship between volume and open interest. Finally the change of lot size and its impact on the causality from open interest to volatility in two different periods indicates firstly the noise trading and secondly hedging does depend on the lot size of the contract.

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